

Module 2 Assignment — Chi Square and ANOVA

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Introduction

In the field of data analytics, the ability to draw meaningful conclusions from data is paramount. Inferential statistics provide the formal framework for this process, allowing analysts to test hypotheses and quantify uncertainty. Among the most fundamental and versatile tools in this domain are the Chi-Square (χ^2) test and the Analysis of Variance (ANOVA). These methods, though both used for hypothesis testing, are applied in distinct scenarios defined by the nature of the data and the research question at hand.

The Chi-Square test is a non-parametric method indispensable for the analysis of categorical data. It serves two primary functions: the Goodness of Fit test, which assesses whether an observed frequency distribution aligns with a theoretical expectation, and the Test of Independence, which examines whether a statistically significant association exists between two or more categorical variables.

In contrast, ANOVA is a parametric technique designed to compare the means of three or more groups. It determines whether observed differences between group means are statistically significant or simply due to random sampling variability. One Way ANOVA evaluates the impact of a single categorical factor on a continuous dependent variable, while Two Way ANOVA expands this analysis to assess the effects of two factors simultaneously, including their potential interaction.

This report will apply these statistical methodologies to a series of analytical problems. Each problem will be systematically addressed using the five-step hypothesis testing procedure. The analyses are conducted using the R programming language, with all results, interpretations, and relevant visualizations presented to provide comprehensive and data-driven solutions.

Analysis of Statistical Problems

The following sections detail the application of Chi-Square and ANOVA tests to various datasets. Each problem is addressed using the traditional five-step hypothesis testing method: (a) stating the null and alternative hypotheses, (b) finding the critical value at a specified significance level, (c) computing the test value, (d) making a statistical decision, and (e) summarizing the results in context.

Chi-Square Goodness-of-Fit Tests

The Goodness-of-Fit test is used to determine if an observed frequency distribution fits an expected theoretical distribution.

Problem 11-1.6: Blood Types

This analysis sought to determine if the blood type distribution at a large hospital ($n=50$) significantly differed from the distribution of the general population at a 0.10 significance level.

- **Hypotheses:**
 - H_0 : The distribution of blood types among hospital patients is the same as in the general population.
 - H_1 : The distribution of blood types among hospital patients is different from that of the general population. (Claim)
- **Critical Value:** With four categories, the degrees of freedom (df) are $4-1=3$. At $\alpha=0.10$, the critical χ^2 value is 6.251.
- **Test Value:** The computed χ^2 test statistic was 5.471.

- **Decision:** Since the test value (5.471) is less than the critical value (6.251), we fail to reject the null hypothesis.
- **Summary:** There is not sufficient statistical evidence at the 0.10 level of significance to conclude that the hospital's blood type distribution is different from that of the general population.

Problem 11-1.8: On-Time Performance by Airlines

This test was conducted to see if a major airline's on-time performance record for 200 flights differed from the national statistics provided by the Bureau of Transportation Statistics at an $\alpha=0.05$ significance level.

- **Hypotheses:**
 - H_0 : The airline's on-time performance distribution is the same as the government's statistics.
 - H_1 : The airline's on-time performance distribution differs from the government's statistics. (Claim)
- **Critical Value:** With $df=3$ and $\alpha=0.05$, the critical χ^2 value is 7.815.
- **Test Value:** The computed χ^2 test statistic was 17.832.
- **Decision:** Since the test value (17.832) is greater than the critical value (7.815), we reject the null hypothesis.
- **Summary:** There is sufficient evidence to conclude that this airline's on-time performance record is significantly different from the national statistics. Specifically, the airline had fewer on-time arrivals and more weather-related delays than expected.

Chi-Square Test of Independence

The Test of Independence is used to determine if there is a significant association between two categorical variables.

Problem 11-2.8: Ethnicity and Movie Admissions

This analysis investigated whether movie attendance by year (2013 vs. 2014) was dependent on ethnicity, tested at the $\alpha=0.05$ level.

- **Hypotheses:**
 - H_0 : Movie attendance by year is independent of ethnicity.
 - H_1 : Movie attendance by year is dependent on ethnicity. (Claim)
- **Critical Value:** With $df=(2-1)(4-1)=3$ and $\alpha=0.05$, the critical χ^2 value is 7.815.
- **Test Value:** The computed χ^2 test statistic was 60.144.
- **Decision:** We reject the null hypothesis, as the test value is substantially larger than the critical value.
- **Summary:** There is strong statistical evidence to support the claim that a relationship exists between ethnicity and movie attendance by year.

Problem 11-2.10: Women in the Military

This test examined whether a relationship exists between rank (Officer vs. Enlisted) and branch of the Armed Forces for women in the military, at $\alpha=0.05$.

- **Hypotheses:**
 - H_0 : Rank is independent of the branch of the Armed Forces.
 - H_1 : A relationship exists between rank and branch of the Armed Forces. (Claim)
- **Critical Value:** With $df=3$ and $\alpha=0.05$, the critical χ^2 value is 7.815.
- **Test Value:** The computed χ^2 test statistic was an overwhelmingly large 654.27.

- **Decision:** The null hypothesis is strongly rejected.
- **Summary:** There is very strong evidence to conclude that a significant relationship exists between a woman's rank and her branch in the Armed Forces, suggesting the proportion of officers to enlisted personnel is not the same across all branches.

One-Way Analysis of Variance (ANOVA)

One-Way ANOVA is used to compare the means of three or more groups.

Problem 12-1.8: Sodium Contents of Foods

This test was performed to determine if a significant difference in mean sodium amounts exists among three food categories: condiments, cereals, and desserts, at $\alpha=0.05$.

- **Hypotheses:**
 - $H_0: \mu_{\text{condiments}} = \mu_{\text{cereals}} = \mu_{\text{desserts}}$.
 - H_1 : At least one mean sodium amount is different. (Claim)
- **Critical Value:** With degrees of freedom $df_{\text{num}}=2$ and $df_{\text{den}}=19$, the critical F-value is 3.52.
- **Test Value:** The computed F-statistic was 2.446.
- **Decision:** Since the test value (2.446) is less than the critical value (3.52), we fail to reject the null hypothesis.
- **Summary:** There is not sufficient evidence to conclude that a significant difference in mean sodium amounts exists among the three food groups based on this sample data.

Two-Way Analysis of Variance (ANOVA)

Two-Way ANOVA examines the effects of two independent factors on a dependent variable.

Problem 12-3.10: Increasing Plant Growth

This analysis tested the effects of two factors, Grow-light strength (2 levels) and Plant Food type (2 levels), on plant growth.

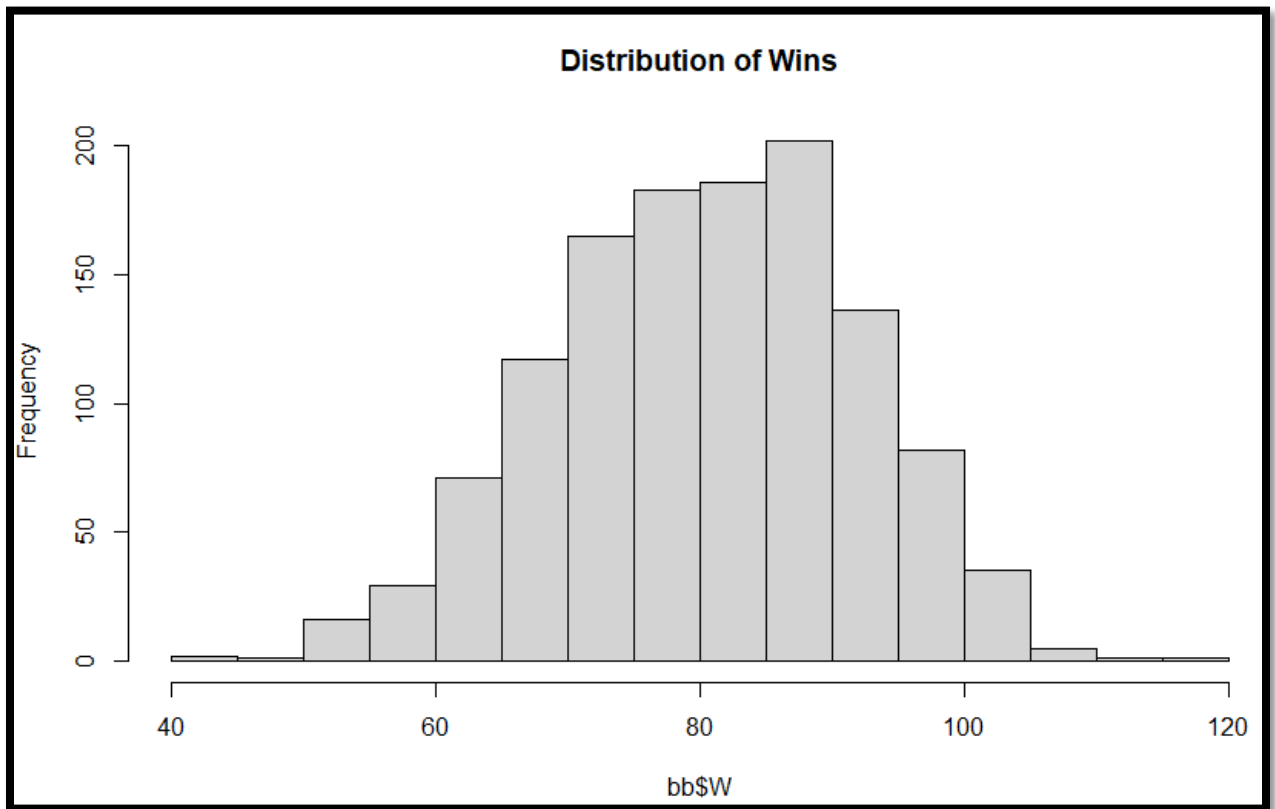
- **Hypotheses:**
 - Interaction ($H_{0,int}$): There is no interaction effect between grow-light and plant food.
 - Main Effect of Light ($H_{0,light}$): There is no difference in mean growth due to light strength.
 - Main Effect of Food ($H_{0,food}$): There is no difference in mean growth due to plant food.
- **Critical Value:** For all tests, $df_{num}=1$ and $df_{den}=8$. At $\alpha=0.05$, the critical F-value is 5.32.
- **Test Values:** The computed F-statistics were: $F_{light} = 3.681$, $F_{food} = 24.562$, and $F_{interaction} = 1.438$.
- **Decision:** We reject the null hypothesis for Plant Food ($24.562 > 5.32$) but fail to reject for Light and the Interaction.
- **Summary:** The type of plant food has a statistically significant effect on plant growth. However, the strength of the grow-light and the interaction between light and food were not found to be significant.

Advanced Analysis in R

Baseball Dataset Analysis:

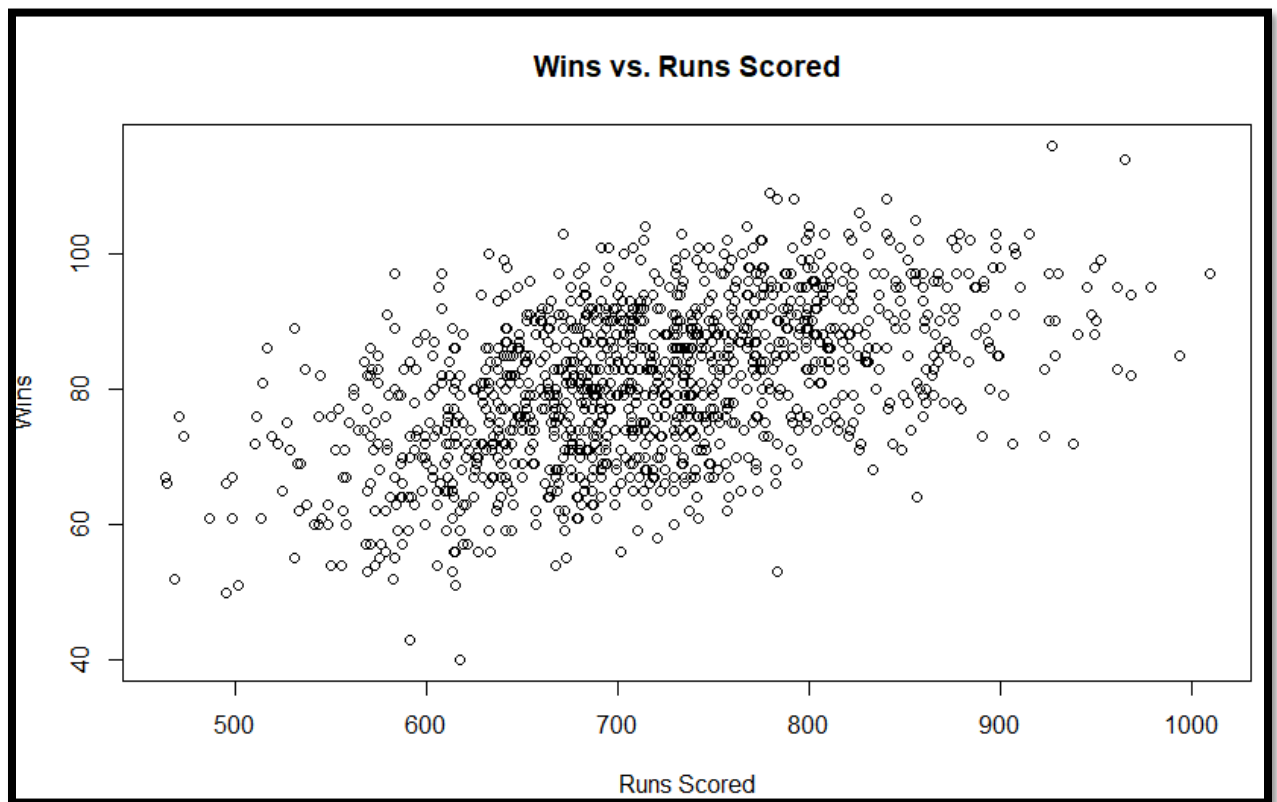
An exploratory data analysis (EDA) was performed on the baseball-1.csv dataset, which contains team statistics from 1962 to 2012. The distribution of team wins (W) per season is approximately normal, centred around a mean of 81 wins.

Figure 1. Distribution of Team Wins per Season.



A scatter plot of Wins versus Runs Scored (RS) showed a strong, positive, linear relationship, confirming that teams that score more runs tend to win more games.

Figure 2. Relationship between Runs Scored and Wins.



A Chi-Square Goodness-of-Fit test was then used to determine if the total number of wins was equally distributed across five decades (1960s-2000s).

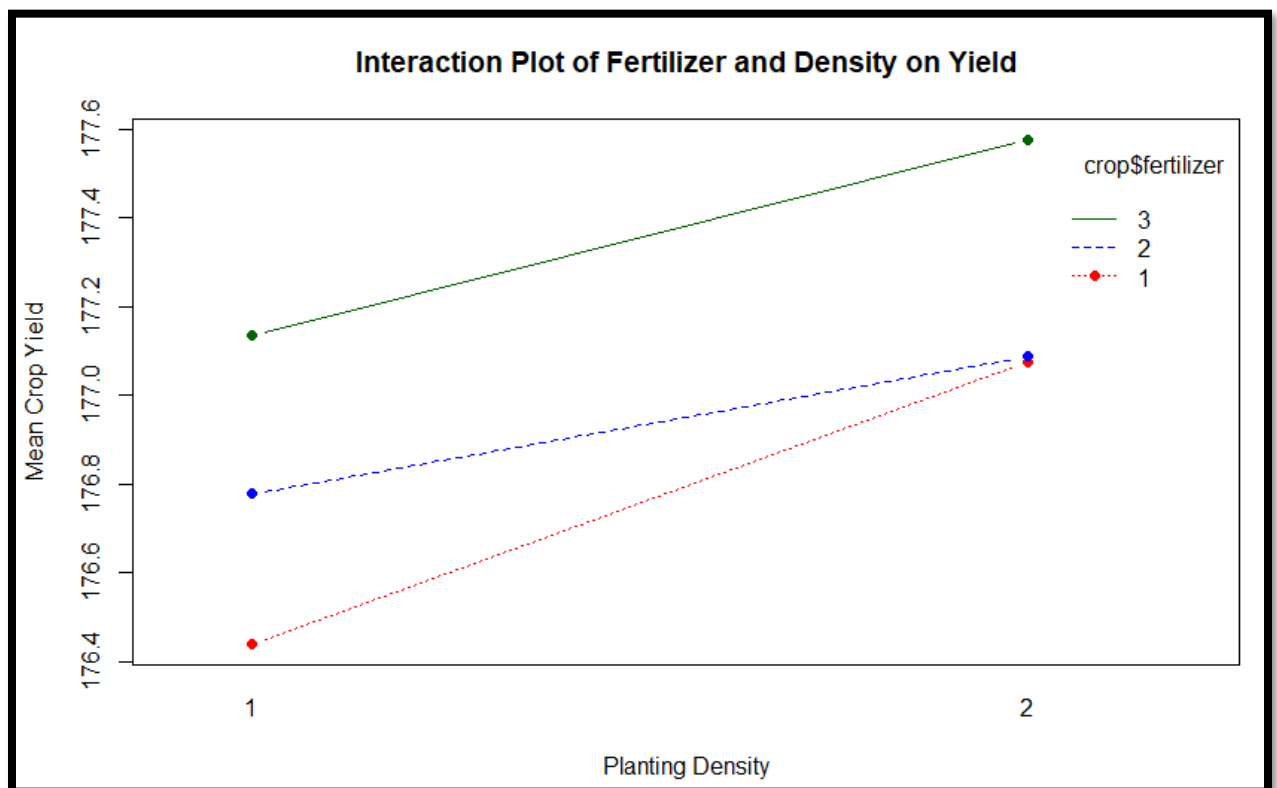
- **Hypotheses:**
 - H0: The number of wins is equally distributed across the decades.
 - H1: The number of wins is not equally distributed across the decades. (Claim)
- **Critical Value:** With $df=4$ and $\alpha=0.05$, the critical value is 9.488.
- **Test Value:** The computed χ^2 statistic was 3336.
- **Decision:** The null hypothesis is rejected. The p-value ($p<.001$) was significantly smaller than $\alpha=0.05$, leading to the same conclusion.

- **Summary:** There is a significant difference in the number of wins across decades, which is likely due to factors such as league expansion (more teams and thus more total wins) over time.

Crop Data Analysis:

A Two-Way ANOVA was performed on the `crop_data-1.csv` dataset to analyze the impact of fertilizer type (3 levels) and planting density (2 levels) on crop yield.

Figure 3. Interaction Plot for Crop Yield.



- **Hypotheses:** Three sets of hypotheses were tested for the main effects of fertilizer, density, and their interaction.

- **Results:** The ANOVA revealed statistically significant main effects for both fertilizer ($F(2,88)=8.944, p<.001$) and density ($F(1,88)=15.099, p<.001$). The interaction effect was not significant ($F(2,88)=0.631, p=.535$).
- **Summary:** Both fertilizer type and planting density are significant factors in determining crop yield. However, because the interaction is not significant, the effect of a particular fertilizer is consistent across different planting densities. The interaction plot above visually supports this conclusion, as the lines representing the different fertilizers are nearly parallel.

Conclusion and Interpretations

This report successfully applied Chi-Square and Analysis of Variance (ANOVA) tests to a diverse set of problems, demonstrating their utility in analysing both categorical and continuous data. The Chi-Square analyses revealed significant relationships where they were hypothesized, such as between military rank and branch, and confirmed when distributions differed from expectations, as with airline performance. In contrast, the test for blood types showed that not all observed differences are statistically significant.

The ANOVA tests provided insights into the factors affecting various outcomes. One-way ANOVA models showed that, based on the sample data, there were no significant differences in means for sodium content in foods, sales across company types, or per-pupil expenditures across regions. The more complex Two-Way ANOVA models highlighted the importance of examining multiple factors simultaneously. For the plant growth experiment, it was found that plant food was the only significant factor. For the crop yield analysis, both fertilizer and density were significant, but their effects were independent of one another.

A structured hypothesis testing framework to avoid drawing conclusions based on intuition alone. The comparison of a test statistic to a critical value, or a p-value to a significance level, provides an objective basis for making decisions. Furthermore, the use of visualizations, such as the scatter plot for baseball wins and the interaction plot for crop yield, proved invaluable for interpreting and communicating the results of the statistical tests. In conclusion, the appropriate application of these fundamental statistical methods provides a powerful lens through which to analyze data, test claims, and uncover meaningful insights that can inform decisions in a wide range of fields.

References

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- R Core Team. (2023). *R: A language and environment for statistical computing*.
R Foundation for Statistical Computing. <https://www.R-project.org/>
- U.S. Department of Commerce. (n.d.). *The New York Times Almanac*.
- U.S. Department of Transportation. (n.d.). *Bureau of Transportation Statistics*.
Retrieved from www.transtats.bts.gov

Appendix: R

```
#Blood Types
```

```
observed_blood <- c(12, 8, 24, 6)
```

```
probs_blood <- c(0.20, 0.28, 0.36, 0.16)
```

```
chisq.test(observed_blood, p = probs_blood)
```

```
# Critical Value
```

```
qchisq(1 - 0.10, df = 3)
```

```
# On-Time Performance
```

```
observed_airline <- c(125, 10, 25, 40) # On time, NAS, Late, Other
```

```
probs_airline <- c(0.708, 0.082, 0.090, 0.120)
```

```
chisq.test(observed_airline, p = probs_airline)
```

```
# Critical Value
```

```
qchisq(1 - 0.05, df = 3)
```

```
#Ethnicity and Movie Admissions
```

```
movie_data <- matrix(c(724, 335, 174, 107, 370, 292, 152, 140), nrow = 2, byrow = TRUE)
```

```
colnames(movie_data) <- c("Caucasian", "Hispanic", "African American", "Other")
```

```
rownames(movie_data) <- c("2013", "2014")
```

```
chisq.test(movie_data)

# Critical Value

qchisq(1 - 0.05, df = 3)


# Women in the Military

military_data <- matrix(c(10791, 7816, 932, 11819, 62491, 42750, 9525, 54344), nrow = 2,
byrow = TRUE)

colnames(military_data) <- c("Army", "Navy", "Marine Corps", "Air Force")

rownames(military_data) <- c("Officers", "Enlisted")

chisq.test(military_data)

# Critical Value

qchisq(1 - 0.05, df = 3)


# 12-1.8. Sodium Contents

condiments <- c(270, 130, 230, 180, 80, 70, 200)

cereals <- c(260, 220, 290, 290, 200, 320, 140, 160)

desserts <- c(100, 180, 250, 250, 300, 360, 300)

sodium_df <- data.frame(

  sodium = c(condiments, cereals, desserts),

  group = factor(rep(c("Condiments", "Cereals", "Desserts"),
```

```
times = c(length(condiments), length(cereals), length(desserts))))))

sodium_aov <- aov(sodium ~ group, data = sodium_df)

summary(sodium_aov)

# Critical Value

qf(1 - 0.05, df1 = 2, df2 = 19)


# 12-2.10. Sales for Leading Companies

cereal <- c(578, 320, 264, 249, 237)

chocolate <- c(311, 106, 109, 125, 173)

coffee <- c(261, 185, 302, 689)

sales_df <- data.frame(

  sales = c(cereal, chocolate, coffee),

  company = factor(rep(c("Cereal", "Chocolate", "Coffee"),

    times = c(length(cereal), length(chocolate), length(coffee))))))

sales_aov <- aov(sales ~ company, data = sales_df)

summary(sales_aov)

# Critical Value

qf(1 - 0.01, df1 = 2, df2 = 11)


# 12-2.12. Per-Pupil Expenditures
```



```
eastern <- c(4946, 5953, 6202, 7243, 6113)
middle <- c(6149, 7451, 6000, 6479)
western <- c(5282, 8605, 6528, 6911)
exp_df <- data.frame(
  exp = c(eastern, middle, western),
  region = factor(rep(c("Eastern", "Middle", "Western"),
    times = c(length(eastern), length(middle), length(western))))))
exp_aov <- aov(exp ~ region, data = exp_df)
summary(exp_aov)
# Critical Value
qf(1 - 0.05, df1 = 2, df2 = 10)

# Increasing Plant Growth
growth <- c(9.2, 9.4, 8.9, 7.1, 7.2, 8.5, 8.5, 9.2, 8.9, 5.5, 5.8, 7.6)
light <- factor(rep(c(1, 1, 2, 2), each = 3))
food <- factor(rep(c("A", "B", "A", "B"), each = 3))
plant_df <- data.frame(growth, light, food)
plant_aov <- aov(growth ~ light * food, data = plant_df)
summary(plant_aov)
# Critical Value
```

```
qf(0.95, 1, 8)

library(dplyr)

bb <- read.csv("baseball-1.csv")

#EDA
summary(bb)
hist(bb$W, main="Distribution of Wins")
plot(bb$RS, bb$W, main="Wins vs. Runs Scored", xlab="Runs Scored", ylab="Wins")

# Chi-Square Test
bb$Decade <- bb$Year - (bb$Year %% 10)
# Filter for full decades & relevant years
bb_filtered <- bb %>% filter(Decade >= 1960 & Decade <= 2000)
wins <- bb_filtered %>%
  group_by(Decade) %>%
  summarize(total_wins = sum(W))
chisq.test(wins$total_wins)

# Critical Value
```

```
qchisq(0.95, df = 4) # 5 decades - 1

crop <- read.csv("crop_data-1.csv")

crop$fertilizer <- as.factor(crop$fertilizer)
crop$density <- as.factor(crop$density)
crop$block <- as.factor(crop$block)

crop_aov <- aov(yield ~ fertilizer * density + block, data = crop)
summary(crop_aov)

interaction.plot(x.factor = crop$density,
                 trace.factor = crop$fertilizer,
                 response = crop$yield,
                 fun = mean, type = "b", pch = 19,
                 main = "Interaction Plot of Fertilizer and Density on Yield",
                 xlab = "Planting Density", ylab = "Mean Crop Yield",
                 col = c("red", "blue", "darkgreen"))
```